Monodromy theory for the Hénon map

Q. What is monodromy?

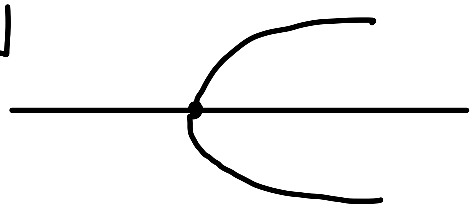
A. Group homomorphism of the form

$\rho : \pi_1(\text{param space} \setminus \text{singular pt}) \to \mathrm{Aut}(\text{"Solutions"})$

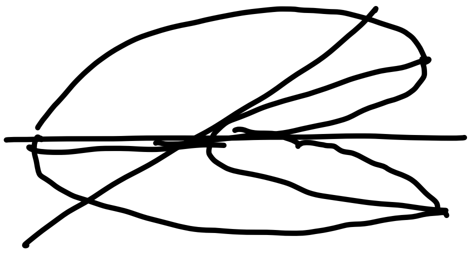
EXAMPLE

$x^2 = c \qquad x = \pm\sqrt{c}$

$\mathbb{R}$



$\mathbb{C}$



Let $c(\theta) = e^{i\theta}$
for $\theta \in [0, 2\pi]$
Sol for $\theta = 0$ are
$x_+ = 1$ & $x_- = -1$,

For each θ, sols are $e^{i\frac{\theta}{2}}$, $e^{i\left(\frac{\theta}{2}+\pi\right)}$.

When θ varies from 0 to 2π,

x_+ continues to x_-

Param sp $= \mathbb{C}\ (\ni c)$ Solutions $= \{x_+, x_-\}$

Singular pt $= 0$ (where two sols collaide)

$\pi_1(\text{param sp} \setminus \text{Singular pt}) \cong \mathbb{Z}$

$\mathrm{Aut}(\text{"Solutions"}) = S_2 = \mathbb{Z}_2$

$\rho : \mathbb{Z} \to \mathbb{Z}_2 \qquad \rho(1) = 1$

Singularity

Outside "singularities" all solutions are uniquely continued w.r.t. parameters.

OTHER EXAMPLES

☆ Galois theory,

☆ Linear ODE over $\mathbb{C}$. $\dot{x} = A(t)x$, $x \in \mathbb{C}^n$

$A(t)$ is hol for $t \in G \subset \mathbb{C}$

$t_0 \in G$: base pt.

Monodromy hom $\pi_1(G, t_0) \longrightarrow GL(n, \mathbb{C})$

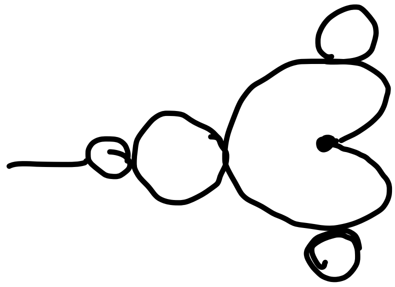
Quadratic Map & Hénon map

$f_c : \mathbb{C} \longrightarrow \mathbb{C}$ $\quad z \longmapsto z^2 + c$

$H_{a,c} : \mathbb{C}^2 \longrightarrow \mathbb{C}^2$ $\quad (x, y) \longmapsto (x^2 + c - ay, x)$

Paramsp	$\mathbb{C} \ni c$	$\mathbb{C}^2 \ni (a, c)$
Singularity	Mandelbrot set	"M"
Solutions	$\Sigma_2^{\mathbb{N}}$	$\Sigma_2^{\mathbb{Z}}$
$\pi_1(\text{Param} \setminus \text{Sing})$	$\mathbb{Z}$	? huge grp.
$\text{Aut}(\text{Sols})$	$\mathbb{Z}_2$	? huge grp.

Quadratic Map



$f_c(z) = z^2 + c$

<u>fixed pts</u> $z = z^2 + c$

$$\iff z = \frac{1 \pm \sqrt{1-4c}}{2}$$

Singularity for the fixed pts $c = \frac{1}{4}$

Stability of ppts

$f : X \to X$ $\quad f^n(x) = f \circ \cdots \circ f(x) = x$

periodic pt of period n

fixed pt of f^n

(if $x \neq f^i(x)$ for $i = 1, \cdots, n-1$
we say x is periodic of prime period n)
(least period)

x : pp of period n

x is transversal $\iff 1 \notin \mathrm{Spec}(Df^n_x)$

x is hyperbolic $\iff \forall \lambda \in \mathrm{spec}(Df^n_x)$
$|\lambda| \neq 1$

transversal $\Rightarrow$ x persists

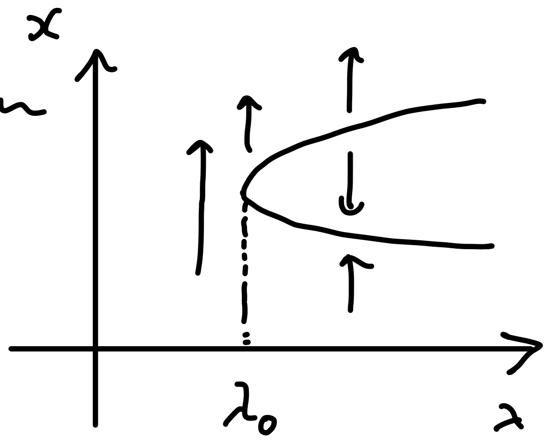
hyperbolic $\Rightarrow$ dynamics on a nbd of x persists

Saddle-Node & Period doubling

$f_\lambda : X \to X$

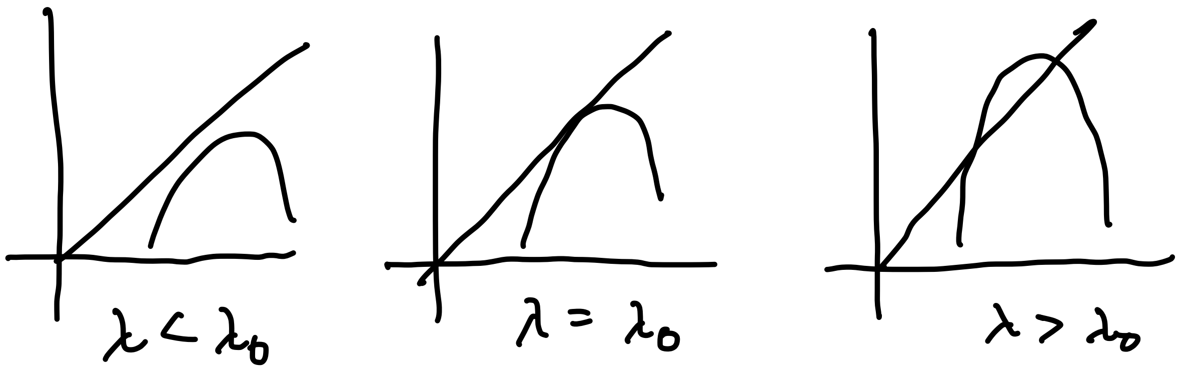
$f_{\lambda_0}(x) = x$

☆ Saddle-Node bifurcation



$Df_{\lambda_0}|_x$ has eigen val 1

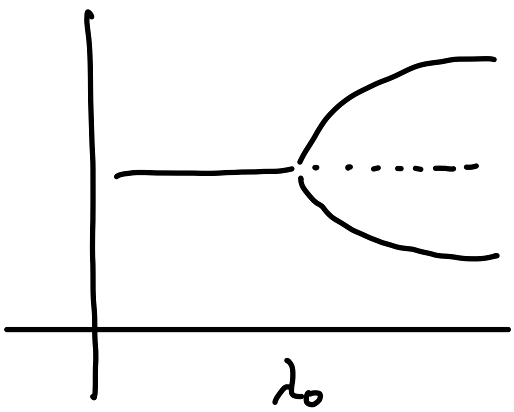
+ generic conditions

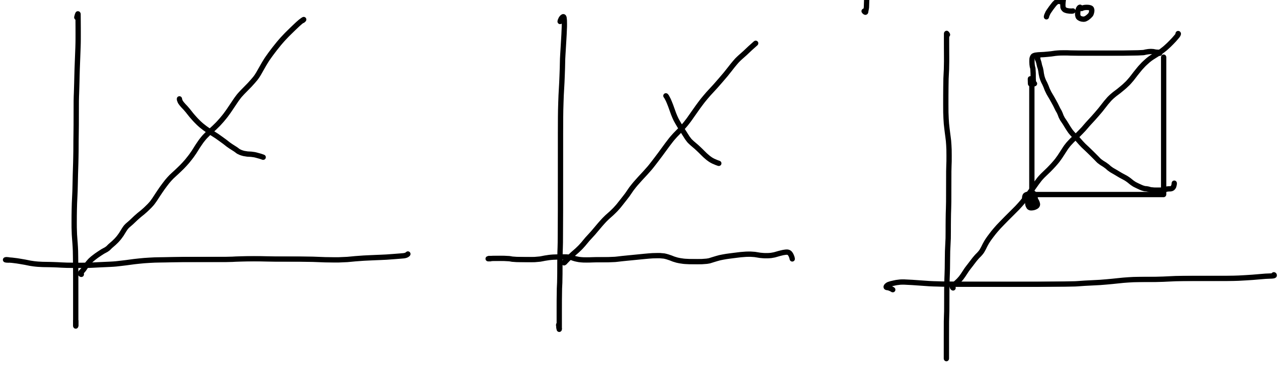


☆ Period doubling bifurcation

$Df_{\lambda_0}|_x$ has eigen val -1

+ generic conditions





Fixed pt of f_c

$$x = \frac{1 \pm \sqrt{1-4c}}{2}$$

$$Df_c\big|_x = 2x = 1 \pm \sqrt{1-4c}$$

x is non-transversal $\Leftrightarrow$ $c = \frac{1}{4}$

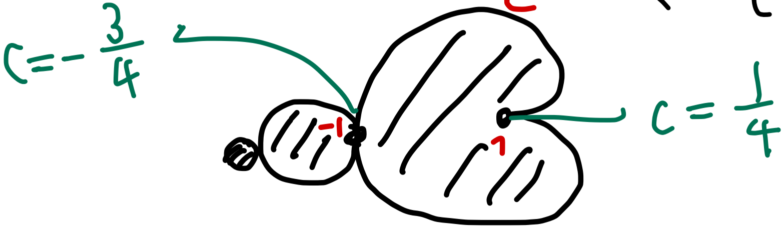
$\Leftrightarrow$ f_c has unique fixed pt.

x goes through period doubling at

$$c = -\frac{3}{4}$$

$$(z^2+c)^2 + c = z$$

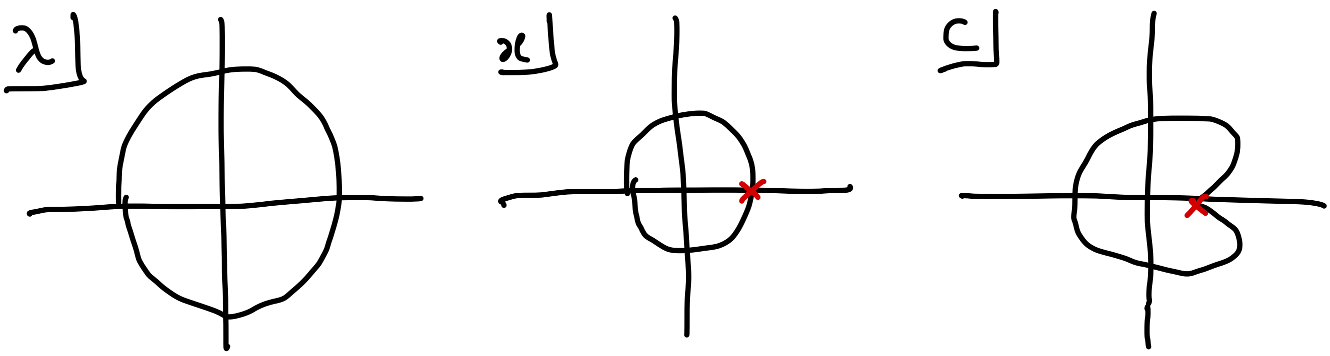
$$\Leftrightarrow (z^2 - z + c)(z^2 + z + 1 + c)$$



Rem

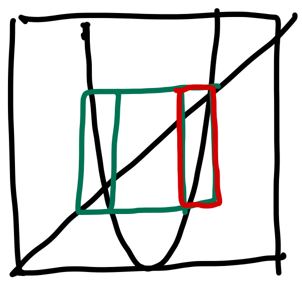
eigen val		fixed pt		param c
λ_p	$\longleftrightarrow$	x_p	$\longrightarrow$	c
λ_q	$\longleftrightarrow$	x_q	$\longrightarrow$	c

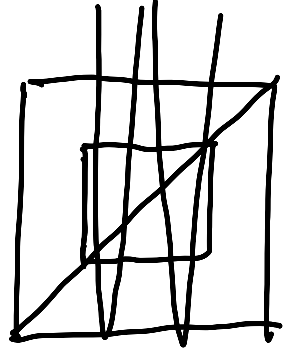
$\lambda = 2x$ $\qquad\qquad$ $c = x - x^2$



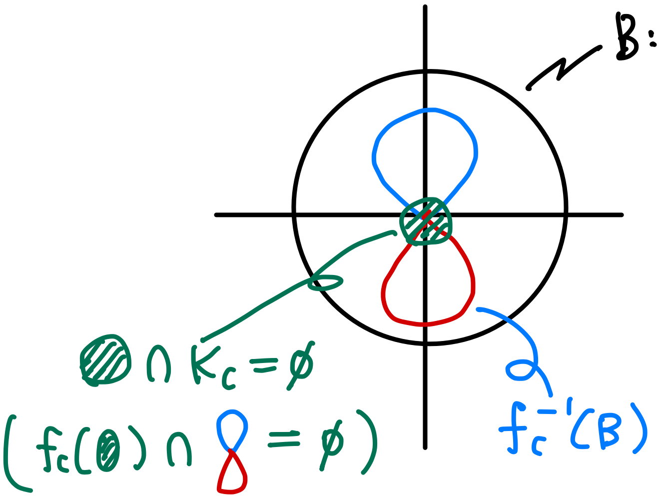
Example $c = -6$

Crit. pt gives a symbolic coding





Example $c = 6$



B: radius 6.

outside of B, $f^n(z) \to \infty$

$$|z^2+6| \geq |z^2| - 6$$
$$\geq |z^2| - |z|$$
$$\geq (|z|-1)|z|$$
$$\geq 5|z|$$

Monodromy Rep

$\gamma \in \pi_1(\mathbb{C} \setminus M)$

$\gamma(0) = \gamma(1)$.

Fix $h : K_{\gamma(0)} \longrightarrow \Sigma_2^{\mathbb{N}}$ and

Construct a conti. fam. $h_t : K_{\gamma(t)} \longrightarrow \Sigma_2^{\mathbb{N}}$

s.t. $h_0 = h$.

$h_1 \circ h_0^{-1} : \Sigma_2^{\mathbb{N}} \longrightarrow \Sigma_2^{\mathbb{N}}$ is an auto.

$\rho : \pi_1(\mathbb{C} \setminus M) \longrightarrow \mathrm{Aut}(\Sigma_2^{\mathbb{N}})$

FACT

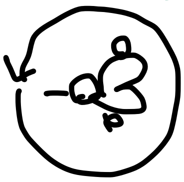
☆ $\mathbb{C} \setminus M \cong \mathbb{C} \setminus \mathbb{D} \Rightarrow \pi_1(\mathbb{C} \setminus M) \cong \mathbb{Z}$

☆ $\mathrm{Aut}(\Sigma_2^{\mathbb{N}}) \cong \mathbb{Z}_2 = \{ id, \phi \}$

ϕ is a 1-block map $\begin{array}{l} 0 \longrightarrow 1 \\ 1 \longrightarrow 0 \end{array}$

involution.

ρ is surjective

γ :  $\rho(\gamma) = \phi$.

∵ Symbolic coding by $f_c^{-1}(\{|z| = 6\})$

interchanges and by γ.

Higher degree Case

THM (Blanchard-Devaney-Keen 1991 Invention Math)

$\rho : \pi_1(S_d, *) \longrightarrow \mathrm{Aut}(\Sigma_d^{\mathbb{N}})$ is surjective

S_d : the param sp for Monic, centered poly of deg d with escaping crit pts.

Hubbard's Conjecture

$$\rho : \pi_1(H, *) \longrightarrow \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$$

H : "shift locus" of the Hénon map

Γ : im ρ.

Conj im ρ and σ generate $\mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$

Theorem

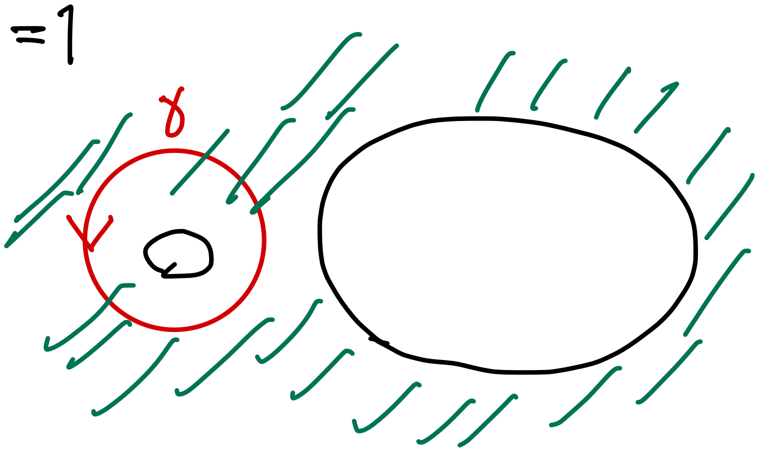
(1) im ρ is non trivial

im ρ contains $\exists$ element of ∞ order

(2) im $\rho \subset \mathrm{inert}(\Sigma_2^{\mathbb{Z}}) \subset \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$

Nontrivial Loops

$b=1$

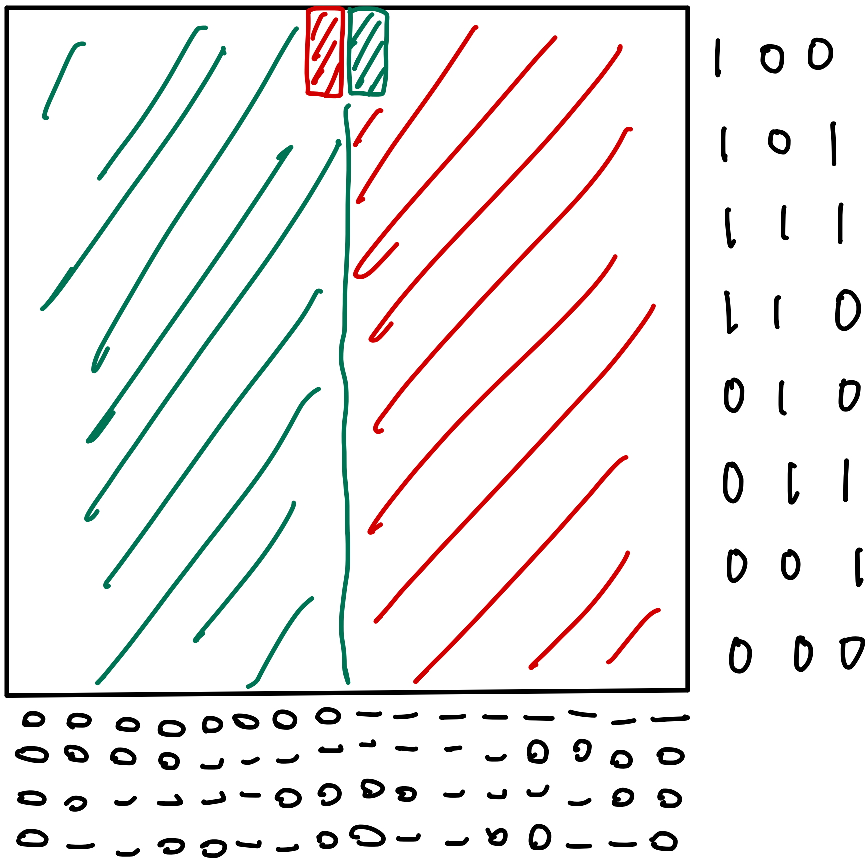


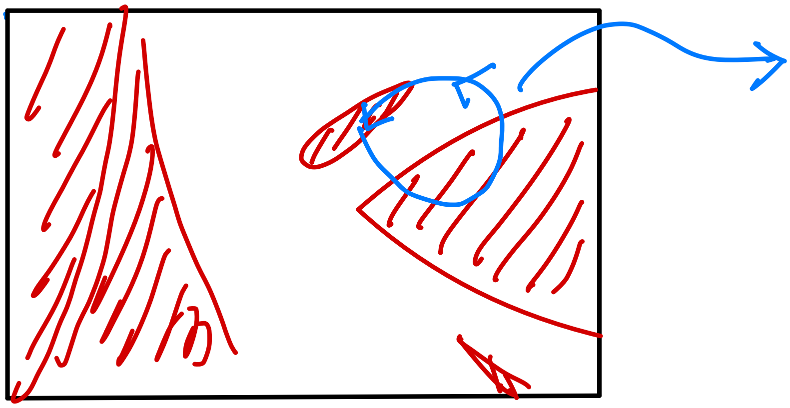
$\rho(\gamma)$ interchanges

0 0 1 0 1 0 0 with

0 0 1 1 1 0 0

involution





This loop interchanges $0010 \longleftrightarrow 0110$

$\alpha : 0 \longleftrightarrow 1$ (big loop around M)

$\beta : 0010 \longleftrightarrow 0110$

<u>Thm</u> order of $\alpha\beta$ is ∞

Proof Let

$$x^{2k} := \cdots 01010110110\,(01)^k.\,1111\cdots$$

$$x^{2k+1} := \cdots 1010110110\,(01)^k\,1.\,0000\cdots$$

then $(\alpha\beta)^n(x^0) = x^n$

$$x^0: \quad \cdots 01010110110.\,1111\cdots$$

$$\beta x^0: \quad \cdots 01010010010.\,1111\cdots$$

$$\alpha\beta x^0: \quad \cdots 10101101101.\,0000\cdots = x^1$$

$$\beta x^1: \quad \cdots 10100100101.\,0000\cdots$$

$$\alpha\beta x^1: \quad \cdots 010110110(10).\,1111\cdots = x^2$$

Monodromy & Real dynamics.

$\mathcal{H}$: hyperbolic horseshoe locus

$* \in \mathcal{H}$ Real base pt.

Assume $(a,c) \in \mathbb{R}^2$ can be connected to $*$ by a path $\alpha \subset \mathcal{H}$.

$\gamma := \overline{\alpha}^{-1} \cdot \alpha$ then

$$\begin{tikzcd}
K^{\mathbb{R}}_{a,c} \arrow[r, "H_{a,c}"] \arrow[d, "h_{\frac{1}{2}}|_{K^{\mathbb{R}}_{a,c}}"', "\cong"] \arrow[dr, phantom, "\circlearrowright"] & K^{\mathbb{R}}_{a,c} \arrow[d, "\cong"', "h_{\frac{1}{2}}|_{K^{\mathbb{R}}_{a,c}}"] \\
\mathrm{Fix}(\rho(\gamma)) \arrow[r] & \mathrm{Fix}(\rho(\gamma))
\end{tikzcd}$$

where $K^{\mathbb{R}}_{a,c} = K_{a,c} \cap \mathbb{R}^2$,

h_t is the family of homeos $K_{a,c} \to \Sigma_2^{\mathbb{Z}}$

$(\gamma(\frac{1}{2}) = (a,c))$

<u>Proof</u> Note that $H_{a,c}$ is symetric wrt cpx conjugate $\phi : (x,y) \mapsto (\overline{x}, \overline{y})$

that is, $\phi \circ H_{a,c} = H_{\overline{a},\overline{c}} \circ \phi$

$\gamma = \bar{\alpha}^{-1} \circ \alpha \qquad \bar{\gamma} = \alpha^{-1} \circ \bar{\alpha}$

$\Rightarrow \quad \bar{\gamma} = \gamma^{-1}$

Take $\forall z \in K^{\mathbb{C}}_{a,c}$ $\quad S_z := h_{\frac{1}{2}}(z) \in \Sigma_2$

We show $\rho(\gamma)(S_z) = S_z \iff z \in \mathbb{R}^2$

$z(\gamma, t) :=$ continuation of z along γ

$= (h_t)^{-1}(S_z)$

☆ $\overline{z(\gamma, t)} = \bar{z}(\bar{\gamma}, t)$

From the symmetry wrt cpx conj

$= \bar{z}(\gamma, 1-t)$

$\Rightarrow \rho(\gamma)(S_z) = h_1(h_0^{-1}(S_z))$

$= h_1(z(\gamma, 0))$

$K^{\mathbb{C}}_{h(0)} \subset \mathbb{R}^2$

$= h_1(\overline{z(\gamma, 0)})$

$= h_1(\bar{z}(\bar{\gamma}, 0))$

$= h_1(\bar{z}(\gamma, 1))$

$= h_1(h_1^{-1}(S_{\bar{z}})) = S_{\bar{z}}$

$\therefore \rho(\gamma)(S_z) = S_z \iff S_z = S_{\bar{z}}$

Since $h_{\frac{1}{2}}$ is homeo, $\iff z = \bar{z}$